Pratik Adhikari

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Research Interests

Power electronics and control for renewable-energy integration, power-system modeling and dynamics, distribution systems, and digital twins and data-driven monitoring and forecasting of power systems.

Education

Institute of Engineering, Pulchowk Campus, Tribhuvan University Apr 2022 – Aug 2026
Bachelor of Engineering in Electrical Engineering Lalitpur, Nepal

- **Aggregate: 84.72%** — Distinction (institutional scale; 80% distinction threshold).
- Final-year project (B.E. thesis): *Development of a Digital Twin for Real-Time Monitoring and Short-Term Forecasting of a Photovoltaic System.*
- Coursework: Power System Analysis I & II, Power Electronics, Control Systems, Digital Control System, Applied Photovoltaic Engineering, Switchgear & Protection, Instrumentation I & II, Engineering Mathematics.

SOS Hermann Gmeiner Secondary School, Gandaki 2019 – 2021
Higher Secondary (Grade 11–12), Science — GPA: 3.64 / 4.00

Research Experience

Digital Twin for Real-Time Monitoring and Short-Term Forecasting of a Grid-Connected PV System Sep 2025 – May 2026
Final-year project (four-member team), Department of Electrical Engineering, Pulchowk Campus. Supervisors: Dr. Basanta Kumar Gautam (Pulchowk Campus) and Er. Deependra Neupane (Electricity Regulatory Commission).

- Built a converter-level MATLAB/Simulink twin of a 335 W grid-connected PV module (single-diode model, DC–DC stage with perturb-and-observe MPPT, single-phase inverter) and validated it against 41 field measurements on a clear and a partly cloudy day.
- Verified MPPT operation against the module's own P–V curve (99.9–100% tracking efficiency, 200–1000 W/m^2). Against the 41 field readings the twin gives 24.2 W RMSE ($r = 0.969$) with on-site irradiance and 99.8 W ($r = 0.69$) with satellite-derived irradiance, showing the value of site-specific sensing. On the clear day, measured power was 87.6% of the twin's prediction, indicative of unmodelled losses.
- Designed and calibrated a low-cost Arduino/LDR irradiance sensor against a smartphone-based reference (power-law fit, $R^2 = 0.974$, $n = 21$) and used it to drive the twin from live serial data in a real-time demonstration.
- Developed a SARIMA$(1,1,1) \times (1,1,1)_{24}$ day-ahead forecaster for irradiance and temperature and propagated it through the twin: on a held-out day, daily energy was within 0.8% of that obtained with measured weather, against 6.7% for a persistence baseline.

Deadbeat Predictive Current Control and Passive Damping of an LCL-Filtered Grid-Connected Converter May 2026 – Present
Simulation study with Panas Bhattarai (Dept. of Electrical Engineering, Tribhuvan University), who provided guidance; I lead the modeling, simulation, analysis and writing.

- Developed a stationary $\alpha\beta$-frame deadbeat predictive current controller for a 4.1 kW, 380 V two-level voltage-source converter with an LCL filter (2.478 kHz resonance): power commands map algebraically to current references, needing no Park transformation or synchronous-frame regulator. Simulated with 10 kHz SVPWM and one control period of computation delay.
- Determined how much passive damping the controller needs: the undamped converter does not operate (voltage limiter active at every step, power flow reverses), no deadbeat gain from 0.05 to 1 restores operation, and the minimum damping ratio lies between 0.034 and 0.057.
- Identified a damping range of 21–30 Ω per delta branch that operates under ±20% L_1, L_2, C tolerance and up to 5 mH added grid inductance, which the 10 Ω minimum does not. Worst harmonic-limit utilisation is 0.34–0.23 (IEC 61000-4-7 subgroups against IEEE 519-2022) at 0.15–0.21% dissipation, about 14 times the fundamental-frequency estimate.

- Earlier study (5 kW, 400 V, no computation delay): 60 Ω per branch removed 91.8% of the resonance-band current; subgroup TDD 3.0–4.9% across six operating points (at most 5.1% over ten robustness cases). Preceded by FCS-MPC and L versus LCL comparisons.

Professional Experience

DeltaQ Engineering — Electrical Engineer (Remote) May 2026 – Present

US-based mechanical, electrical and plumbing (MEP) consultancy

- Perform electrical load calculations and system analysis for residential, multi-family, commercial and industrial projects in accordance with applicable US codes and standards.
- Prepare AutoCAD single-line diagrams, specifications and coordination layouts; check equipment selections.

Honors & Awards

Ranked 3rd in the graduating Electrical Engineering cohort Aug 2026

Merit Scholarship, Institute of Engineering, Tribhuvan University Apr 2022 – Aug 2026

- Awarded on the national engineering entrance exam (approx. 10,000 candidates); IOE stipend, all eight semesters.

Engineering Design Projects (Undergraduate)

16 MW Run-of-River Hydropower Plant Design 2026

- Hydrology, turbine, generator and transformer selection, grounding, station supply, backup sizing.

132 kV Double-Circuit Transmission Line Design 2026

- 108 MW over 220 km: voltage level, tower geometry, insulation design, conductor evaluation.

Commercial Building Power Distribution & Illumination 2025

- AutoCAD layouts, board/cable sizing, protection coordination, 250 kVA transformer, 330 A MCCB.

Smart Fertigation System 2023

- Arduino irrigation control (DHT-11, soil-moisture sensors) with prototype fertilisation control.

Technical Skills

Simulation: MATLAB/Simulink (Simscape Electrical), DIgSILENT PowerFactory (coursework), COMSOL Multiphysics, Proteus
Methods: predictive current control, SVPWM, LCL filter design and damping, harmonic analysis (IEC 61000-4-7, IEEE 519), MPPT, time-series forecasting (SARIMA), model verification
Programming & data: Python (NumPy, SciPy, pandas, statsmodels, scikit-learn, Matplotlib), MATLAB, C/C++, JavaScript
Hardware: Arduino, sensor calibration, serial data acquisition, PCB design
Tools: LaTeX, Git, AutoCAD, Bluebeam Revu

Outreach

BloomED — STEM Content Creator (Volunteer) Jun 2021 – Jan 2022

- Wrote simplified science and technology learning material for under-resourced students.

Test Scores

IELTS Academic (1 Nov 2026): Overall **8.0** (Listening 9.0, Reading 9.0, Writing 7.0, Speaking 7.0)

Languages

Nepali (native), English (fluent), Hindi (conversational)